

# Literature Review

## Paper 1

### AlmaHub : An Engaging, Supportive Alumni-Students Interaction Platform

#### Authors

Sachin S Patil  
Atharv Bhasme  
Pranav Bobade  
Abhishek Barkade  
Pranav Pore

#### Publication Site

[2023 IEEE 8th International Conference for Convergence in Technology \(I2CT\), Pune, India](#)

DOI: 10.1109/I2CT57861.2023.10126226

#### Overview

This paper introduces *AlmaHub*, a digital interaction platform developed to strengthen the relationship between alumni and current students within an educational institution. The system aims to create a supportive ecosystem where alumni can remain connected with their institution and contribute to student development. By providing a structured online platform, the system facilitates continuous engagement and improves communication efficiency between stakeholders.

#### Methodology

The platform is implemented as a centralized web-based application where users are required to register and authenticate before accessing system features. Separate access roles are provided for alumni and students to manage interactions appropriately. The system architecture focuses on ease of use and secure access, ensuring that interactions occur within a controlled digital environment.

| Merits                                                         | Demerits                                                            |
|----------------------------------------------------------------|---------------------------------------------------------------------|
| Enhances alumni–student interaction in a structured manner     | System usefulness depends on consistent user participation          |
| Provides a dedicated platform for engagement and communication | Functional scope is primarily limited to interaction-based features |
| Supports relationship building within academic institutions    |                                                                     |

# Paper 2

## Alumni Management and Networking System

### Authors

Rajini S

Hari Prasad B

Upendrasingh A

### Publication Site

[2023 2nd International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation \(ICAECA\)](#)

**DOI:** 10.1109/ICAECA56562.2023.10200060

### Overview

This paper presents an alumni management and networking system designed to store alumni data and facilitate connections among graduates. The primary goal of the system is to provide institutions with a centralized mechanism to maintain alumni records and promote professional networking among former students.

### Methodology

The system is developed as a centralized application with modules for alumni data storage and networking. Alumni details are collected and managed through the system, enabling users to access and update information. Networking features allow alumni to connect with peers through the platform.

| Merits                                               | Demerits                                        |
|------------------------------------------------------|-------------------------------------------------|
| Centralized management of alumni records             | Security mechanisms are not discussed in detail |
| Facilitates networking among alumni                  | Scalability considerations are limited          |
| Reduces reliance on informal communication platforms |                                                 |

# Paper 3

## Design of Alumni Portal with Data Security

### Authors

Sankara Eshwari S

Ramya Chitra

Sandhiya K

### Publication Site

[2021 Second International Conference on Electronics and Sustainable Communication Systems \(ICESC\)](#)

**DOI:** 10.1109/ICESC51422.2021.9532986

### Overview

This paper focuses on the design of an alumni portal with a strong emphasis on protecting sensitive alumni data. The system aims to address concerns related to unauthorized access and data misuse by incorporating security-oriented design principles into the alumni portal.

### Methodology

The proposed portal includes user authentication and access control mechanisms to ensure that only authorized users can access specific information. Secure handling of alumni records is prioritized throughout the system design, ensuring data confidentiality and integrity.

| Merits                                  | Demerits                                                 |
|-----------------------------------------|----------------------------------------------------------|
| Strong focus on data security           | Limited emphasis on communication or networking features |
| Controlled access to alumni information | Functional scope is primarily security-oriented          |
| Improves system reliability and trust   |                                                          |

# Paper 4

## Alumni Mentorship Portal Using Cloud Computing

### Authors

A Induja  
Rani Robsi  
B Latha

### Publication Site

[2022 1st International Conference on Computational Science and Technology \(ICCST\)](#)

**DOI:** 10.1109/ICCST55948.2022.10040315

### Overview

This paper proposes a cloud-based alumni mentorship portal designed to connect alumni with students for guidance and mentoring purposes. The system aims to leverage cloud computing to ensure accessibility and availability of mentorship services across locations.

### Methodology

The portal is deployed using cloud infrastructure, enabling users to access the system through the internet. Alumni and students register on the platform and interact for mentorship-related activities. Cloud deployment ensures scalability and centralized system management.

| Merits                                     | Demerits                                            |
|--------------------------------------------|-----------------------------------------------------|
| Supports structured alumni mentorship      | System performance depends on internet connectivity |
| Cloud-based access improves availability   | Data privacy aspects are not extensively discussed  |
| Reduces dependency on local infrastructure |                                                     |

## Paper 5

# An Efficient System for Interconnecting Alumni After Their Studies

### Authors

Rohit Rastogi  
Shweta Singh  
Rashmi Singh  
Shrishti Garg  
Gautam Yadav

### Publication Site

[2023 5th International Conference on Advances in Computing, Communication, Control and Networking \(ICAC3N\), Greater Noida, India](#)

**DOI:** 10.1109/ICAC3N60023.2023.10541457

### Overview

This paper presents an online system aimed at interconnecting alumni after the completion of their academic programs. The system focuses on maintaining long-term connections among alumni and enabling continuous association with the institution.

### Methodology

The system is designed as a centralized online platform where alumni can register and interact. The architecture supports data storage and user interaction to ensure that alumni remain connected even after graduation.

| Merits                                      | Demerits                                                    |
|---------------------------------------------|-------------------------------------------------------------|
| Maintains long-term alumni connectivity     | User engagement is critical for system effectiveness        |
| Centralized platform for alumni interaction | Limited discussion on advanced features or system expansion |
| Supports institutional alumni networks      |                                                             |

## Paper 6

# Dynamic Alumni Monitoring with Decision Support System

### Authors

Marifel Grace Capili-Kummer

Maria Leodevina C. Batugal

### Publication Site

[International Journal of Recent Technology and Engineering \(IJRTE\).](#)

[Volume 9, Issue 5, January 2021](#)Published by [Blue Eyes Intelligence Engineering and Sciences Publication](#)

**DOI :** 10.35940/ijrte.F5308.019521

### Overview

This paper presents the design and development of a **web-based Dynamic Alumni Monitoring System integrated with a Decision Support System (DSS)**. The primary goal of the system is to collect **real-time data on graduate employability**, track alumni professional advancement, and support university administrators in data-driven decision-making. The study emphasizes the importance of alumni tracer studies for curriculum relevance, accreditation compliance, and employability assessment. The system was developed and implemented at St. Paul University Philippines to address challenges in alumni traceability and reporting .

### Methodology

The researchers adopted an **Agile software development methodology**, specifically **Extreme Programming (XP)**. The development process followed six phases: exploration, planning, iterations to release, production, maintenance, and death. Data were collected through alumni questionnaires and administrative consultations. The system was evaluated using **ISO/IEC 25010 Software Quality Standards**, assessing functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Statistical tools such as weighted mean and frequency distribution were used to analyze compliance results

## **Merits**

Enables **real-time monitoring of alumni employability**

Integrates **decision support features** for administrators

Reduces manual, paper-based alumni tracking

Complies with **ISO/IEC 25010 software quality standards**

Provides automated reporting for institutional planning

Improves communication between alumni and university

## **Demerits**

Primarily **web-based**, lacks dedicated mobile application support

Depends on **alumni participation** for accurate data collection

Requires continuous system maintenance and updates

Initial setup and adoption may require administrative training

Limited discussion on advanced analytics or predictive modeling

## Paper 7

# Development of Web-Based Tracer Alumni Information System

### Authors

Lidya Rosnita

Yesy Afrillia

Rizky Putra Fhonna

Ulva Ilyatin

### Publication Site

[Journal of Computer Science, Information Technology and Telecommunication Engineering \(JCoSITTE\)](#)

[Volume 2, Number 2, September 2021, Pages 202–210](#)

Published by Universitas Muhammadiyah Sumatera Utara

**DOI :**[10.30596/jcositte.v2i2.7845](#)

### Overview

This paper presents the development of a web-based tracer alumni information system aimed at improving graduate tracking after completion of studies. The study addresses the problem of manual alumni data collection, which was found to be inefficient and prone to data management issues at the Faculty of Social and Political Sciences, Malikussaleh University. The proposed system enables structured storage of alumni biodata and employment information, supports tracer studies, and assists institutions in fulfilling accreditation requirements. The system also strengthens alumni–institution relationships by providing a centralized digital platform for alumni data management.

### Methodology

The authors adopted a field study and literature study approach to analyze the existing alumni data collection process. Based on the analysis, a web-based system was designed using structured system analysis techniques.

- System modeling was carried out using Flowcharts, Context Diagrams, Data Flow Diagrams (DFD), and Entity Relationship Diagrams (ERD).



- The application was implemented using PHP with XAMPP and MySQL for database management.
- Two user roles were defined: Admin and Alumni, with role-based access control for data entry, updates, and reporting.
- The system was tested to ensure accurate data storage, retrieval, and report generation

### **Merits**

- Automates alumni tracer studies and eliminates manual data collection
- Provides structured and centralized alumni data storage
- Supports alumni employment and profile tracking
- Improves efficiency and accuracy of data management
- Assists institutions in meeting accreditation and reporting requirements
- Enhances alumni–institution communication

### **Demerits**

- Limited to a web-based system without mobile application support
- Does not include decision support or advanced analytics features
- Relies on alumni participation for data accuracy
- Implemented as a single-institution case study, limiting generalization
- Requires continuous system maintenance and administrative oversight

## Paper 8

# Development of Intelligent Alumni Management System for Universities

### Authors

Nayyar Ahmed Khan

Ahmed Masih Uddin Siddiqi

Mohammad Ahmad

### Publication Site

[Asian Journal of Basic Science & Research \(AJBSR\),  
Volume 3, Issue 2, Pages 51–60, April–June 2021](#)

**DOI:** 10.38177/ajbsr.2021.3206

### Overview

This paper presents the design and implementation of an **intelligent, web-based Alumni Management System (AMS)** aimed at strengthening communication between universities, graduates, and employers. The study highlights the need for continuous interaction with alumni to understand labor market demands, improve curriculum quality, and enhance graduate employability. The proposed system provides a centralized digital platform where alumni can manage profiles, share skills, search for job opportunities, and maintain professional and social connections with their institution after graduation.

### Methodology

The system was developed using a **software engineering life cycle approach**, including analysis, design, implementation, testing, and deployment phases.

- Requirements were identified for three main user groups: alumni, university administrators, and employers.
- System modeling was carried out using **UML diagrams**, including use case diagrams, activity diagrams, sequence diagrams, and ER diagrams.
- The application was implemented as a **web-based system** using **ASP.NET, C#, SQL Server/MySQL**, and hosted on an IIS web server.

- Functional and non-functional requirements such as usability, security, performance, and maintainability were considered.
- System testing was conducted to ensure accuracy, security, and reliability before deployment.

### **Merits**

- Provides a **centralized platform** for alumni information management
- Improves communication between alumni, universities, and employers
- Supports alumni profile management, CV uploads, and job searching
- Enables employers to search for qualified graduates securely
- Enhances curriculum evaluation through alumni feedback
- Focuses on usability, security, and accessibility

### **Demerits**

- Limited to a **web-based application** with no mobile app integration
- Lacks advanced analytics or decision-support features
- Depends heavily on alumni participation for data accuracy
- No real-time tracking or automated reporting mechanisms
- Scalability to multiple institutions is not fully discussed

# Paper 9

## Tracer Study Information System for Higher Education

### Authors

P. W. Yunanto

A. Idrus

V. M. Santi

A. S. Hanif

### Publication Site

[IOP Conference Series: Materials Science and Engineering,](#)

[Volume 1098, Paper No. 052107, 2021](#)

[Proceedings of the 5th Annual Applied Science and Engineering Conference \(AASEC 2020\)](#)

Published by IOP Publishing

**DOI:**10.1088/1757-899X/1098/5/052107

### Overview

This paper presents the development of a **Tracer Study Information System** designed to support higher education institutions in monitoring graduates after completion of their studies. The study emphasizes that universities have a responsibility not only to educate students but also to ensure the sustainability of graduates' careers. The proposed system acts as a **centralized, one-door platform** for collecting alumni employment data and using it as feedback to improve curriculum design, student development programs, and institutional policies. The system was developed as a case study for Universitas Negeri Jakarta (UNJ)

### Methodology

The system was developed using the **Waterfall Software Development Life Cycle (SDLC)** model.

- The development stages included **analysis, design, implementation, testing, and maintenance**.
- Functional and non-functional system requirements were identified during the analysis phase.

- System modeling was carried out using **Use Case Diagrams and Activity Diagrams**.
- The application was implemented using **PHP** as the programming language and **MySQL** as the database.
- System testing was conducted using **black-box testing**, focusing on functional suitability as per **ISO/IEC 25010** quality standards.
- Test results showed that all system functions operated as expected on a local server environment

### **Merits**

- Provides a centralized tracer study platform for universities
- Supports systematic alumni employment data collection
- Helps improve curriculum, policies, and student skill development
- Uses standard software engineering and testing practices
- Aligns with accreditation and quality assurance requirements

### **Demerits**

- Implemented as a single-university case study
- Limited to a web-based system without mobile application support
- Does not include advanced analytics or decision-support features
- Requires continuous alumni participation for complete data coverage
- System testing was conducted only in a local environment

# Paper 10

## Development of Web-Based Tracer Alumni Information System

### Authors

Lidya Rosnita

Yesy Afrillia

Rizky Putra Fhonna

Ulva Ilyatin

### Publication Site

[Journal of Computer Science, Information Technology and Telecommunication Engineering \(JCoSITTE\).](#)

[Volume 2, Number 2, September 2021, Pages 202–210](#)

Published by Universitas Muhammadiyah Sumatera Utara.

**DOI:** 10.30596/JCOSITTE.V2I2.7845

### Overview

This paper focuses on the development of a **web-based tracer alumni information system** aimed at improving the tracking of graduates after completion of their studies. The study highlights that alumni data collection at the Faculty of Social and Political Sciences, Malikussaleh University was previously conducted manually, making it inefficient and error-prone. The proposed system facilitates structured storage of alumni data, supports graduate employment tracking, and assists institutions in meeting accreditation requirements. The system is designed to strengthen alumni–institution relationships and provide valuable feedback for academic evaluation.

### Methodology

The authors employed **field studies and literature studies** to analyze existing alumni data collection processes. Based on the findings, a web-based system was designed using **structured system analysis techniques**.

- System modeling was carried out using **Context Diagrams, Data Flow Diagrams (DFD), and Entity Relationship Diagrams (ERD)**.

- The system was implemented using **PHP with XAMPP** and **MySQL** as the database.
- Admin and alumni roles were defined to manage access control, data entry, and report generation.
- System testing ensured proper data validation, storage, and retrieval before deployment

### **Merits**

- Replaces manual alumni data collection with an automated web-based system
- Enables structured and centralized alumni data storage
- Supports alumni employment and profile tracking
- Improves efficiency and accuracy of data management
- Assists institutions in accreditation and quality assurance processes
- Enhances alumni–institution communication

### **Demerits**

- Limited to a **web-based platform** with no mobile application support
- Lacks advanced analytics or decision-support features
- Depends on alumni willingness to update their information
- Implemented as a **single-faculty case study**, limiting generalization
- Requires continuous technical maintenance and administrative support

# 11.Automated Alumni Data Tracking System

**Published in:** [\*International Journal of Health Sciences\*](#)

**Year:** 2022

**Authors:** Babu, P., Deepadharshini, G., Priya, S. D., & Janani, A.

## Overview

This paper proposes an **internet-based Automated Alumni Data Tracking System** designed to help educational institutions maintain accurate and up-to-date alumni records. The system is implemented using a **web portal and Android application**, allowing alumni to update their profiles, locations, and professional details. It centralizes alumni information and enables institutions to track graduate activities, career progression, and geographical distribution efficiently.

The system also supports **notifications and announcements**, improving alumni–institution engagement and ensuring long-term connectivity after graduation.

## Key Features

- **Centralized Alumni Database:** Stores academic, professional, and contact details in a single platform
- **Self-Profile Updating:** Alumni can independently update career and location data
- **Multi-Platform Access:** Web-based portal with Android application support
- **Notification System:** Enables institutions to send updates, events, and announcements
- **Location Tracking:** Helps institutions understand alumni distribution globally

| merits                                   | Demerits                                     |
|------------------------------------------|----------------------------------------------|
| Eliminates manual alumni data collection | Depends on alumni willingness to update data |
| Improves long-term alumni engagement     | Limited analytical and predictive features   |
| Reduces data redundancy and loss         | Requires continuous system maintenance       |
| Enables easy communication with alumni   | Privacy concerns if security is weak         |

**DOI:** 10.53730/ijhs.v6nS8.9978



# 12.Alumni Network Platform Leveraging Regression Models for Data Analysis

**Published in:** African Journal of Biological Sciences (Afr.J.Bio.Sc.)  
**Volume:** 6, Issue Si2, 2024  
**ISSN:** 2663-2187

**Authors:**Nolan M. Yumen, MIS

## Overview

This paper presents the design and implementation of a **web-based alumni network platform** for the University of Antique – Tario-Lim Memorial Campus (UA-TLMC). The system addresses challenges in alumni data management, communication, and long-term engagement by replacing manual and fragmented record-keeping methods with a centralized digital solution.The study follows a **descriptive developmental research approach** involving 300 respondents (students, faculty, alumni, and novice users). Using **regression analysis**, the research identifies key factors influencing user satisfaction, such as ease of navigation, clarity of information, responsiveness, and mobile accessibility. The platform integrates data analytics to generate insights into alumni employability trends, engagement patterns, and institutional impact, thereby supporting strategic decision-making and lifelong alumni relationships.

## Key Features

- **Centralized Alumni Database:** Secure storage and management of alumni personal, academic, and professional data
- **Regression-Based Data Analytics:** Identifies predictors of user satisfaction and alumni engagement
- **Communication & Networking Modules:** Forums, announcements, messaging, and event participation
- **User-Centric Design:** Focus on usability, clarity, and intuitive navigation
- **Mobile Accessibility:** Responsive design for access across devices
- **Security Measures:** Password hashing, session management, and data privacy protection

| Merits                                                               | Demerits                                               |
|----------------------------------------------------------------------|--------------------------------------------------------|
| Eliminates manual and fragmented alumni data handling                | Requires continuous data updates from alumni           |
| Enhances alumni engagement and long-term institutional relationships | Initial development and maintenance cost               |
| Enables data-driven decision making through analytics                | Dependent on user participation for accuracy           |
| Improves communication efficiency and transparency                   | Limited advanced AI features in current implementation |

**DOI:**10.33472/AFJBS.6.Si2.2024.2051-2603

# 13.Development of a Web-Based Platform for Alumni Employability and Career Tracking: A Tracer Study at St. Paul University Philippines (SPUP)

**Published in:** European Journal of Interdisciplinary Social Sciences (EJISS)  
**Year / Issue:** 2025, Vol. 1, Issue 3

**Author:**Carlos Babaran Jr.

## Overview

This paper presents a web-based alumni tracer study platform developed for St. Paul University Philippines (SPUP) to improve graduate employability tracking and alumni engagement. The system addresses the limitations of manual data collection and static alumni webpages that restrict effective career monitoring and institutional decision-making. Developed using a descriptive-developmental approach with the Agile Scrum methodology, the platform was evaluated using the ISO/IEC 25010 Software Quality Model. Results from IT experts and administrators indicate that the system satisfies all eight quality characteristics to a very great extent, confirming its usability, security, and overall effectiveness.

## Key Features

- **Web-Based Tracer Study System:** Centralized tracking of alumni employability and career progression
- **ISO/IEC 25010 Quality Compliance:** Evaluated across usability, performance efficiency, security, and maintainability
- **Agile Scrum Development:** Iterative and flexible system development approach
- **Employment Data Management:** Collection and analysis of graduate employment information
- **Alumni Engagement Modules:** News updates, event announcements, and communication tools
- **Administrative Dashboard:** Supports data-driven academic and institutional planning

| Merits                                                              | Demerits                                                |
|---------------------------------------------------------------------|---------------------------------------------------------|
| Eliminates manual and fragmented alumni tracking processes          | Requires regular alumni participation for accurate data |
| Enhances employability monitoring and institutional decision-making | Initial system setup and training effort                |
| Ensures software quality through ISO/IEC 25010 evaluation           | Limited AI-based analytics in current version           |
| Improves alumni-university communication and engagement             | Dependent on internet connectivity                      |

**DOI:**10.59324/ejiss.2025.1(3).03

# 14. Employability Tracer Study of Information Technology Education Graduates from a State University in the Philippines

**Published in:** Social Sciences & Humanities Open  
**Volume / Year:** Vol. 2, 2020

**Authors:** Albert C. Albina, Lilian P. Sumagaysay

## Overview

This study investigates the **employability of Bachelor of Science in Information Technology (BSIT) and Bachelor of Science in Computer Science (BSCS) graduates** from a state university in the Philippines. The research aims to assess employment status, job relevance, time taken to secure first employment, salary range, and curriculum relevance in relation to graduates' first jobs. A **cross-sectional retrospective survey method** was employed, involving **177 graduates (86.76% response rate)** from batches 2015–2017. Data were collected using the **standard Graduate Tracer Study (GTS) questionnaire** developed by the Commission on Higher Education (CHED). Results show that **78.53% of graduates were employed**, with **69.78% reporting that their first job was related to their degree program**. Most graduates secured employment within **1–2 years**, and the majority perceived the university curriculum as relevant to their work. The study highlights the importance of curriculum alignment and continuous program review to enhance graduate employability.

## Key Features

- **Graduate Tracer Study Approach:** Uses CHED-standardized tracer study instrument
- **Employability Assessment:** Evaluates employment status, job relevance, salary, and job search duration
- **Curriculum Relevance Analysis:** Measures alignment between academic training and industry requirements
- **Competency Evaluation:** Identifies key skills useful in employment, including communication and IT skills
- **Evidence-Based Curriculum Improvement:** Provides data to guide academic and career development programs

| Merits                                                                      | Demerits                                                             |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| Provides empirical evidence on the employability of BSIT and BSCS graduates | Limited to graduates from a single state university                  |
| Demonstrates strong alignment between curriculum and industry relevance     | Relies on self-reported data from graduates                          |
| Supports curriculum review and academic quality assurance                   | Cross-sectional design does not capture long-term career progression |
| Identifies key employability skills such as communication and IT skills     | Lacks predictive or advanced analytical models                       |

**DOI:**10.1016/j.ssaho.2020.100055

# 15.Tracer Study Information System for Higher Education

**Published in:** IOP Conference Series: Materials Science and Engineering  
**Volume / Year:** Vol. 1098, 2021

**Authors:**P. W. Yunanto,A. Idrus,V. M. Santi,A. S. Hanif

## Overview

This paper presents the development of a **one-door web-based Tracer Study Information System** for higher education institutions, with **Universitas Negeri Jakarta (UNJ)** as the case study. The system aims to improve graduate quality by providing structured feedback for curriculum development, student activities, and institutional policy-making. The study highlights that alumni responsibility extends beyond graduation and emphasizes career sustainability and employability tracking. The system was developed using the **Waterfall Software Development Life Cycle**, covering analysis, design, implementation, testing, and maintenance phases. It was implemented using **PHP and MySQL** and evaluated through **black-box functional testing**. Results indicate that all system functions operated as expected, demonstrating that the platform is reliable and suitable for centralized tracer study management at the university level.

## Key Features

- **Centralized Tracer Study System:** One-door alumni tracking managed at the university level
- **Multi-User Access Control:** Supports university admin, study program admin, alumni, stakeholders, and guests
- **Curriculum Feedback Mechanism:** Provides data for curriculum improvement and policy decisions
- **Web-Based Survey Module:** Enables alumni to submit tracer study questionnaires online
- **Functional Testing Validation:** System reliability verified through black-box testing
- **Technology Stack:** Developed using PHP with MySQL database

| Merits                                                            | Demerits                                               |
|-------------------------------------------------------------------|--------------------------------------------------------|
| Centralizes alumni tracer study data across the university        | Limited evaluation to functional testing only          |
| Improves curriculum development and institutional decision-making | Performance and security testing not fully implemented |
| Reduces fragmented, program-level tracer studies                  | Depends on alumni participation for data completeness  |
| Supports accreditation and quality assurance requirements         | Maintenance phase not yet executed                     |
| Simple and structured system design using Waterfall model         | Lacks advanced analytics or predictive features        |

**DOI:**10.1088/1757-899X/1098/5/052107